

Week 5 Handouts — Technical Infrastructure & Modeling

Participant-facing reference cards and fillable templates for Week 5. Each is extracted from a daily lab so triads have the card in hand rather than buried in the activity packet. Distribute the day's set before that day.

Week 5 produces the **TMD** (Technical Modeling Document), Sections 1–5, one section per day, integrated and signed off on Friday.

Day 1 — Database Structures & Data Logic (TMD Section 1)

Handout	What it is
tmd-template	The full 5-section TMD skeleton with [Day X] tags, section-relationship notes + Friday rubric (build it all week)
db-vocabulary-card	The nine data-modeling terms + TPM signal for each (reference)
storage-choice-buckets	The four storage families and when to reach for each (reference)
access-pattern-sheet	Fill-in sheet for every read/write, with frequency + freshness (template)
entity-model-template	Per-entity block: fields, keys, indexes, relationships, invariants (template)
trade-off-template	Option A/B / Choice / Why / Cost / Trigger — carried from Week 4 (template)

Day 2 — Cloud Architecture & Infrastructure (TMD Section 2)

Handout	What it is
cloud-vocabulary-card	Eight cloud terms + the five topology decisions (reference)
managed-vs-self-test-card	Managed-first default + four-question test + fill-in table (template)
multi-tenancy-spectrum	The four tenancy stances and how to choose (reference)
rom-cost-table	Rough-order-of-magnitude cost table + reference bands (template)

Day 3 — REST & SOAP API Fundamentals (TMD Section 3)

Handout	What it is
rest-method-map-card	Five HTTP methods, idempotency, status codes (reference)
five-api-aspects	The five aspects a TPM must read in an API contract + REST-vs-SOAP (reference)
openapi-endpoint-template	Shape for documenting one endpoint in detail (template)
idempotency-card	The three mechanisms + per-endpoint fill-in table (template)
versioning-strategies	Three strategies, the default, and a choose-and-document block (reference)
standard-error-codes	The seven most-needed codes + standard error-body shape (reference)

Day 4 — High-Level Technical Modeling (TMD Section 4)

Handout	What it is
sequence-diagram-elements-card	Diagram elements, right-tool guidance, the three sequences needed (reference)

Day 5 — Performance, Monitoring & AI Validation (TMD Section 5)

Handout	What it is
baselines-template	Baselines table + the two kinds of baseline (template)
monitoring-three-layer-reference	On-call / operator / executive layers and chart caps (reference)
alert-design-template	Four-question alert block (template)
ai-summary-validation-log	Cumulative validation log + status values (template)
friday-tmd-rubric	The six-dimension scoring rubric for the Friday cross-review (reference)

Design principle: these are *cards and blank templates*, not answer keys. Worked examples and answer keys stay in the lab activity packets. Templates ship blank so triads produce the analysis themselves.